Self-Organized Criticality: Sandpile Model on Undirected Graph, Influence of Topology and Dissipation Rules on Avalanches

1. Introduction

Natural systems with high degree of freedom such as earthquakes often exhibit a tendency to reach a critical state. This phenomenon is called self-organized criticality. These systems most famously need no fine-tuning in order to reach the critical state, rather it is an inherit property of theirs. Systems in a critical state are very sensitive to inputs and this leads to its unpredictability.

The concept of SOC was first discovered in 1987 by Bak, Tang and Wiesenfeld and its property demonstrated on a sandpile model [1]. This model was later generalized from square lattice to arbitrary graph. That generalization is important because it allows us to study the models behaviour for different graph topologies.

2. Background

2.1. Sandpile Model

The original sandpile model presents a simple chessboard and a falling sand grains. Each grain fall on some chess square determined by a random distribution. If the number of grains on any square reach four, the sandpile topples and the four grains fall on adjoint squares, possibly causing additional squares to topple. In case the square does not have four adjacent squares (ie it is on the edge of chessboard) the remaining grains fall off the board and are therefore removed from the model. (TODO dissipation here???)

This model can be expressed as an undirected graph, a square lattice of the size 8×8. This however is not a correct representation until an additional vertex is introduced. The often called sink vertex have an important property that it cannot be toppled. When the sink is connected to every vertex on the edge of the square lattice (twice to corner vertices) the graph representation becomes valid. This is foundational observation, for it allows us to use any underlying graph for the sandpile model.

TODO: define avalanche

TODO: sink is needed for the model to not end-up in infinite loop - dissipation TODO: what is dissipation and why is it needed (as before) TODO: at least one vertex connected to sink needed (or on random graph one connection to the sink for every vertex is needed)

2.2. Self-Organized Criticality

TODO: critical state, power-law

2.3. Graph Theory

In order to explore the behaviour of the sandpile model on a graph, we must estabilish these concepts. First is *vertex degree* $d(v_i)$ which is defined as equal to the number of vertices connected to v_i via an edge.

Path legth describes the distance between two vertexes for a given path. The path length is equal to the number of vertices in between the first and last vertex on a path. The *shortest path length* refers to a distance between two vertexes such as any other path between them produces at least the same path length. It's useful to also define an *average shortest path length*, an average of shortest paths between any two vertexes in a graph.

TODO: small-world network properties (the previous needed for that)

3. Model Definition

3.1. Dissipation Rules

As discussed in *Sandpile Model (page 1)* the existence of a sink vertex is necessary for any graph representation of a sandpile model. However there are various rules by which you can determine if a given vertex should be connected to a sink vertex and how many times if so.

For example the original model uses a rule (named by me as) *Fill to Four*. This rule states that the degree of vertex must be exactly four. The sink vertex is connected to all vertices as many times is needed to fulfill that rule. For the square lattice, only vertices on the edge are connected (exactly once except for corner vertices, which are connected twice). This rule precisely reproduces the original chessboard idea.

For an arbitrary graph though, the rule is unsufficient. It doesn't ensure dissipation which can lead to graph supersaturation and never ending avalanche. This rule doesn't reflect on the vertex degree if that is higher than four.

One possible rule which guarantee dissipation on any graph may be *All Once*. The rule is fair in a sense that all vertices have connection to the sink and is uniform because every vertex, no matter its degree, is connected exactly once.

Finally another rule which assure dissipation is *As Many As Neighbours*. This rule connects every vertex to sink proportionally to the number of its neigbouring vertices, therefore emphasizing how many connections it has.

Every rule represents a different approach and not all can be applied to all types of graph topology.

3.2. Underlaying Graphs

4. Observed Behaviour

TODO: how did it work for different types of graphs and dissipation rules? also the graph configurations (eg β for WS)

Bibliography

[1] P. Bak, C. Tang, and K. Wiesenfeld, "Self-organized criticality: An explanation of the 1/f noise," *Physical Review Letters*, vol. 59, no. 4, pp. 381–384, July 1987, doi: 10.1103/PhysRevLett.59.381.